7.3 Classification of Welding processes

1. Arc welding
   a. Carbon arc welding
   b. Metal Arc welding
   c. Metal inert gas welding
   d. Tungsten inert gas welding
   e. Atomic hydrogen welding
   f. Plasma arc welding
   g. Submerged welding
   h. Electro slag welding
2. Gas Welding
   a. Oxy – acetylene welding
   b. Air – acetylene welding
   c. Oxy hydrogen welding
3. Welding related processes
   a. Oxy – acetylene cutting
   b. Arc cutting
   c. Hard facing
   d. Brazing
   e. Soldering

7.4 Arc welding

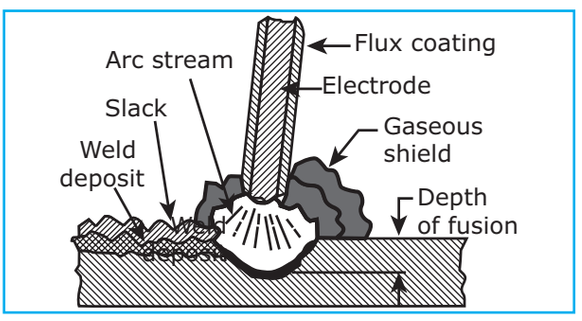
WAS1K6

In arc welding, the edges of two metal parts are melted by an electric arc and the joint is made. An electrode made of a suitable metal is utilised for this purpose. The electrode is taken closer to the parts to be joined and electric current is supplied to both the parts and the electrode. An electric arc is made between the electrode and the metal parts. This arc generates high temperature and melts the metal parts. The parts are joined at this molten state. The filler metal in the form of electrode is deposited along the joint. The metal parts are joined without the application of any pressure. Electrical energy is converted into heat energy in arc welding.





The distance between the metal parts and the electrode should be around 3mm. The heat generated during arc welding ranges from 5000^0C to 6000^0 C. A generator or a transformer supplies the required current to both the electrode and the metal parts. The electrodes are flux coated to prevent the molten metal from reacting with the atmosphere.



Arc welding

7.5 Arc welding equipments

The following equipments are used for the process of arc welding

XII BME_English version CHAPTER 07.indd 129 1/8/2020 1:40:24 PM